

NGUYEN TAN DAT

BACKEND INTERN CANDIDATE

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ABOUT ME

Software Engineering student with a focus on backend architecture, API design, and database systems. I approach engineering as a way to understand how components interact, from clean state flow to predictable logic. Enjoy exploring different areas of software development through hands-on projects, and aspire to become a well-rounded Software Engineer with a strong focus on backend development. Ready to learn CI/CD and established software development practices such as Agile, while working collaboratively and continuously improving through feedback.

EDUCATION

University of Information Technology, VNU-HCM
Bachelor of Software Engineering - GPA:8.53

Sep. 2024 – Jun. 2028
Ho Chi Minh City, Vietnam

AWARDS

DEVO – 20th Anniversary SE Dev Challenge
Second Place – Game Category

2026
UIT-VNU-HCM

SKILLS

Languages: Java, Go, C#

Backend: Spring Boot, REST API, WebSocket, JWT, OAuth2

Databases: PostgreSQL, MongoDB, Redis

Testing: JUnit, Spring Test, JMeter

Tools: Docker, Git, GitHub Actions, Swagger, Postman, Flyway

PROJECTS

Mosquizto | GitHub

Mar. 2026 – Jun. 2026

Learning platform with quiz, flashcard, social, and recommendation features, developed with Android and Spring Boot.

- **Tech stack:** Java, Spring Boot, Android, Redis, WebSocket, Meilisearch, Qdrant, Docker
- **API & Caching:** Designed RESTful APIs and used Redis HSET structures to cache frequently accessed data and reduce repeated database round-trips.
- **Real-time & Authentication:** Built WebSocket notifications with an offline queue; implemented JWT authentication, Google OAuth2, and token-based role grouping for authorization.
- **Search & Recommendation:** Integrated Meilisearch for fuzzy search and built a Qdrant/ONNX recommendation pipeline based on users' learning history.
- **Testing:** Wrote unit tests for core business logic and designed JMeter scenarios to validate API behavior and performance.

Bommy | GitHub

Mar. 2026 – Jul. 2026

Multiplayer game project exploring server lifecycle management, lobby systems, and host-authoritative networking with Unity and PurrNet.

- **Tech stack:** C#, Unity, PurrNet, State Machine, Networking
- **Server Lifecycle:** Modeled the match-server lifecycle as a state machine (Requested -> Ready -> InMatch -> Settling -> Released -> Failed) backed by a shared allocation record.
- **Lobby System:** Designed server-side validation for host-only actions, room-code/password gating, lobby state checks, and host-leave handling.
- **Multiplayer Networking:** Implemented host-authoritative networking with PurrNet using SyncVars and RPCs, separating client input from authoritative gameplay state.